BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: PR XII (002) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** Mathematics Paper–XII (A-02) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. $\int_0^2 x^2\,dx$ = ...

(A) 4/3
(B) 16/3
(C) 8/3
(D) None of these

Q2. $f(x,y) = x^3y + xy^2$ then f_y = ...

(A) $3x^2 + y^2$
(B) $3x^2y + y^2$
(C) $3x^2 + 2xy$
(D) $x^3 + 2xy$

Q3. $\lim_{x\to\infty} (3x^2 + 2)/(x^2 + 1)$ =

(A) ∞
(B) 0
(C) 1
(D) 3

Q4. $\log_e 1$ =.....

(A) 1
(B) e
(C) 0
(D) 10

Q5. If $f(x) = (4x - 3)^3$ then $f'(x)$ = ...

(A) $3(4x - 3)^2$
(B) $12(4x - 3)^2$
(C) $4(4x - 3)^2$
(D) $-12(4x - 3)^2$

Q6. d/dx log_a x =.....

(A) 1/(x lna)
(B) x/lna
(C) (1/x) lna
(D) 1/x

Q7. The distance between two points (3, −2) and (−1, −5) is.....

(A) 10
(B) 5
(C) 15
(D) 20

Q8. The midpoint between two points (a, −b) and (−a, b) is.....

(A) (a/2, b/2)
(B) (2a, 2b)
(C) (0, 0)
(D) (−2a, −2b)

Q9. Two lines are said to be perpendicular if the product of their slopes is equal to.....

(A) 0
(B) −1
(C) None of these

Q10. $y = mx \pm \sqrt{(a^2m^2 + b^2)}$ are equations of.....

(A) Chords
(B) Tangents
(C) Normal
(D) None of these

Q11. Centre of the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ is.....

(A) (4, 6)
(B) (−4, −6)
(C) (−2, −3)
(D) (2, 3)

Q12. In standard form the equation $16y^2 - 9x^2 = 144$ of hyperbola can be written as.....

(A) $y^2/9 - x^2/16 = 1$
(B) $y^2/9 - x^2/16 = 144$
(C) $x^2/16 - y^2/9 = 1$
(D) $y^2/9 + x^2/16 = 1$

Q13. dy/dx = (y − x)/(y + x) is a differential equation of degree.....

(A) 1
(B) −1
(C) 0
(D) None of these

Q14. lim_{x→0} (e^x − 1)/x = ...

(A) e
(B) 0
(C) 1
(D) −1

Q15. The critical values for $f(x) = x^3 - 3x^2 - 9x + 1$ are.....

(A) x = 0, 3
(B) x = −1, 3
(C) x = −1, −3
(D) x = 0, −3

Q16. ∫a^x dx = ...

(A) a^x lna
(B) x^a lna
(C) a^x/lna
(D) None of these

Q17. If f(x) = e^{4x}, then the fifth derivative of the function is.....

(A) [illegible in source scan]
(B) [illegible in source scan]
(C) [illegible in source scan]
(D) [illegible in source scan]

Q18. The graph of the function is concave upward on an open interval (a,b) if

(A) f'(x) > 0
(B) f''(x) < 0
(C) f''(x) > 0
(D) f''(x) ≤ 0

Q19. The Domain of the function F(t) = [illegible in source scan]

(A) R − {−1}
(B) R − {0}
(C) R − {1}
(D) None of these

Q20. ∫tan x dx = ...

(A) ln |sinx| + C
(B) −ln |sinx| + C
(C) ln |cos x| + C
(D) −ln |cos x| + C

Section-B

Time Allowed: 3 Hours 40 Minutes | **Maximum Marks:** 50

Note: Note: Time allowed for Section – B and Section – C is 3 Hours and 40 minutes.

QII. Answer any TEN parts. Each part carries FIVE marks.

1. Evaluate: Lim_{x→0} (sec x − 1)/(x sec x)
2. Find the value of x if $\log_{10}(x + 6) - \log_{10}(x - 3) = 1$
3. If U = $\sqrt[3]{(1 - 3t^2)}$, then find du/dt.
4. If y = $\sin^3(\pi x^3)$ then find y''
5. Find the Machlaurrin's series expansion for f(x) = coshx
6. Find the integration of ∫ $\cos^2$x/cosec x dx by method of substitution.
7. Differentiate the scalar function g(x) = | sinxi − 2xj + cos xk |
8. Find the centroid of the triangle ABC, whose vertices are A(1, 1), B(−2, −2) and C(4, 5).
9. Find the equation of the circle with centre at (0, 5) and circumference passes through the point (5, 0).
10. Find the tangent equation to the parabola y^2 = x which makes an angle 45° with x-axis.

11. For what value of "C" the line $x - y + C = 0$ will touch the ellipse $x^2/4 + y^2/1 = 1$

12. Solve the initial value problem $dy/dx = x - 1$, $y(0) = 1$

13. Find the partial derivative fx and fy for $f(x,y) = x^3 + x^2y + xy^2 + y^3$

Section-C

Time Allowed: 3 Hours 40 Minutes | **Maximum Marks:** 30

> Note: Note: Time allowed for Section – B and Section – C is 3 Hours and 40 minutes.

Q. Attempt any THREE questions. Each question carries equal marks.

QIII.

(a) Find y'' of the function $b^2x^2 + a^2y^2 = a^2b^2$

(b) Evaluate: $\int_1^2 (2x^2 - 3)\, dx$

QIV.

(a) Find the condition that the line $x + y + n = 0$ touches the circle $x^2 + y^2 = 9$

(b) If $v = 2i - j + 5k$ and $\omega = i + 2j - 3k$. Find $d^2/dt^2.(v.t^4\omega)$

QV.

(a) Find the equation of a line that passes through the points (3, 1) and (−1, 3)

(b) Evaluate: $\int_0^1 (x - 1)/(x^2 - 2x + 3)\, dx$

QVI.

(a) Verify Euler's theorem for the function $z = f(x,y) = (x^2 + xy + y^2)^{-1}$

(b) Find dy/dx if y = 5^(x²−x)